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Glossary

Atrial natriuretic peptide – is a polypeptide hormone released by atrial myocytes (muscle cells) from the granules of the atria of the heart in response to high blood pressure, hypervolemia and exercise. It is involved in the homeostatic control of body water and sodium.

Aquaporins – or water channels are formed by specific plasma membrane proteins in the tubular cells. These water channels of the proximal convoluted tubules are always open accounting for the high water permeability in this region. In contrast the water channels in the distal convoluted tubule are regulated by the hormone vasopressin accounting for the variable water re-absorption in that region.

Bowman's capsule hydrostatic pressure – The pressure exerted by the fluid in the Bowman's capsule. This pressure tends to push fluid out of Bowman's capsule, opposes the filtration of fluid from the glomerulus into Bowman's capsule.

Cortical nephrons – All nephrons originate in the cortex, but the glomeruli of the cortical nephrons lie in the outer layer of the cortex. Peri tubular capillaries do not form vasa recta.

Glomerular filtration – this is the first step in urine formation where 20% of the plasma that enters the glomerulus is filtered. The glomerular filtrate that comes out of the glomerulus into the Bowman's capsule is the protein – free plasma.

Glomerular capillary pressure – It is the fluid pressure exerted by the blood within the glomerular capillaries.

Glomerulus – a tuft of capillaries that filters protein – free plasma into the tubular component

Juxtaglomerular apparatus – The ascending limb of Henle returns to the glomerular region of its own nephron, where it passes through the fork formed by the afferent and efferent arterioles. Both the tubular and vascular cells at this point are specialized to form juxtaglomerular apparatus that lie next to the glomerulus. (Juxta means "next to").

Juxtamedullary nephrons – the glomeruli of the juxtaglomerular nephrons lie in the inner layer of cortex next to the medulla and the loops of Henle plunges through the entire depth of the medulla Concentrated urine is formed in these nephrons.

Filtration slits - The narrow slits between adjacent foot process that provides a pathway through which the fluid leaving the glomerular capillaries can enter the lumen of Bowman's capsule.

Peritubular capillaries -supply the renal tissue, involved in exchanges with the fluid in the tubular region.

Podocytes – The glomerular membrane consists of octopus like cells called podocytes that entangles the glomerular tuft. Each podocyte bears many foot processes.

Vasa recta – (straight vessels) The peritubular capillaries of the juxtaglomerular nephrons forms vascular loops which run in close association with the loops of Henle.

Evaluation

1. Arrange the following structures in the order that a drop of water entering the nephron would encounter them.
   a. Afferent arteriole
   b. Bowman's capsule
   c. Collecting duct
   d. Distal tubule
   e. Glomerulus
   f. Loop of Henle
   g. Proximal tubule
   h. Renal pelvis